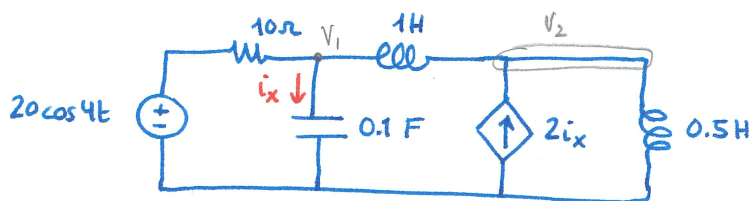


EXERCISE 1 (Nodal analysis)



Find i_x

- ① Go from time domain to phasor domain

$$Z_{10\Omega} = 10 \Omega$$

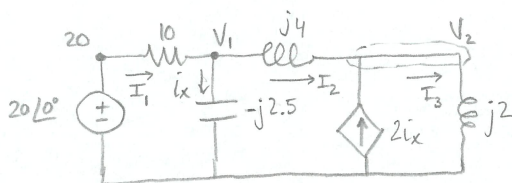
$$Z_{1H} = j\omega L_1 = j4 \times 1 = j4 \Omega$$

$$Z_{0.1F} = \frac{1}{j\omega C} = -j\frac{1}{4 \times 0.1} = -j2.5 \Omega$$

$$Z_{0.5H} = j\omega L_2 = j4 \times 0.5 = j2 \Omega$$

$$V_g = 20 \cos 4t \rightarrow V_g = 20 \angle 0^\circ \text{ V}$$

- ② Use nodal analysis to find V_1 and V_2



KCL @ node 1:

$$I_1 = i_x + I_2$$

$$\frac{20 - V_1}{10} = \frac{V_1}{-j2.5} + \frac{V_1 - V_2}{j4} \Rightarrow 2 - \frac{V_1}{10} = j\frac{V_1}{2.5} - j\frac{V_1}{4} + j\frac{V_2}{4}; \left(\frac{1}{10} + j\frac{1}{2.5} - j\frac{1}{4}\right)V_1 + j0.25V_2 = 2$$

$$(0.1 + j0.15)V_1 + j0.25V_2 = 2 \quad \text{①}$$

KCL @ node 2:

$$I_2 + 2i_x = I_3$$

$$\frac{V_1 - V_2}{j4} + 2\left(\frac{V_1}{-j2.5}\right) = \frac{V_2}{j2} \Rightarrow -j\frac{V_1}{4} + j\frac{V_2}{4} + j\frac{2V_1}{2.5} = -j\frac{V_2}{2}; \left(-j\frac{1}{4} + j\frac{2}{2.5}\right)V_1 + \left(j\frac{1}{4} + j\frac{1}{2}\right)V_2 = 0$$

$$j0.55V_1 + j0.75V_2 = 0 \quad \text{②}$$

$$\begin{cases} \text{①} & (0.1 + j0.15)V_1 + j0.25V_2 = 2 \\ \text{②} & j0.55V_1 + j0.75V_2 = 0 \quad (\times (-1/3)) \end{cases} \Rightarrow \begin{cases} 0.1 + j0.15V_1 + j0.25V_2 = 2 \\ -j0.183V_1 - j0.25V_2 = 0 \end{cases}$$

$$(0.1 - j0.0333)V_1 = 2 \Rightarrow V_1 = \frac{2}{0.1 - j0.0333} = \frac{2 \angle 0^\circ}{0.11 \angle -18.43^\circ}$$

$$V_1 = 18.18 \angle 18.43^\circ \text{ V}$$

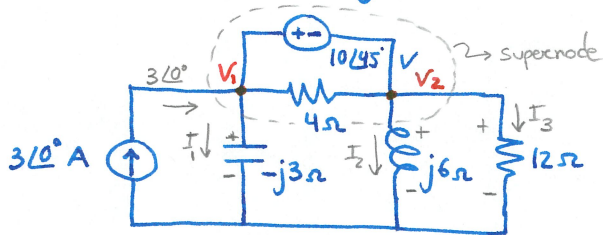
- ③ Calculate I_x and convert back to time domain to find i_x

$$I_x = \frac{V_1}{-j2.5} = j\frac{V_1}{2.5} = j\frac{18.18 \angle 18.43^\circ}{2.5} = \frac{1 \angle 90^\circ \times 18.18 \angle 18.43^\circ}{2.5} = 7.27 \angle 108.4^\circ \text{ A}$$

$$i_x(t) = \underline{\underline{7.27 \cos(4t + 108.4^\circ) \text{ A}}}$$

EXERCISE 2 (Nodal analysis)

Calculate V_1 and V_2



① KCL @ supernode :

$$3\angle 0^\circ = I_1 + I_2 + I_3$$

$$3\angle 0^\circ = \frac{V_1}{-j3} + \frac{V_2}{j6} + \frac{V_2}{12}$$

$$j\frac{1}{3}V_1 + \left(\frac{1}{12} + \frac{1}{j6}\right)V_2 = 3\angle 0^\circ$$

$$j\frac{1}{3}V_1 + \left(\frac{1}{12} - j\frac{1}{6}\right)V_2 = 3\angle 0^\circ$$

② Constraint equation

$$V_1 - V_2 = 10\angle 45^\circ$$

$$\left. \begin{aligned} j\frac{1}{3}V_1 + \left(\frac{1}{12} - j\frac{1}{6}\right)V_2 &= 3 \\ V_1 - V_2 &= 10\angle 45^\circ \quad (\times (-j/3)) \end{aligned} \right\} \begin{aligned} j\frac{1}{3}V_1 + (0.0833 - j0.1666)V_2 &= 3 \\ -j\frac{1}{3}V_1 + j\frac{1}{3}V_2 &= 10\angle 45^\circ \times (-j/3) \end{aligned}$$

$$(0.0833 - j0.1666 + j0.3333)V_2 = 3 + \underbrace{10\angle 45^\circ \times \frac{1}{3}\angle -90^\circ}_{3.333\angle -45^\circ}$$

$$(0.08333 + j0.1666)V_2 = 3 + (2.36 - j2.36)$$

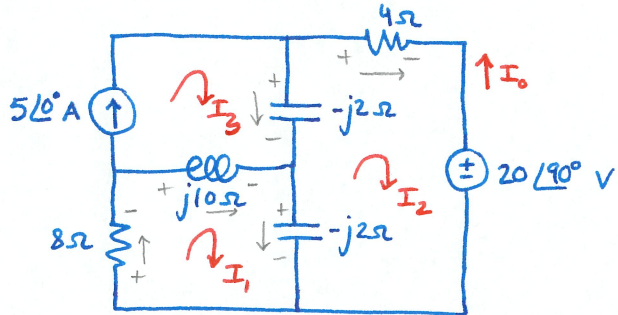
$$V_2 = \frac{5.36 - j2.36}{0.08333 + j0.1666} = \frac{5.86\angle -23.76}{0.19\angle 63.43} = \underline{\underline{30.84\angle -87.19^\circ}}$$

From ② :

$$V_1 - V_2 = 10\angle 45^\circ \Rightarrow V_1 = V_2 + 10\angle 45^\circ = 30.84\angle -87.19^\circ + 10\angle 45^\circ = (1.51 - j30.8) + (7.07 + j7.07)$$

$$\underline{\underline{V_1 = 8.58 - j23.73 = 25.23\angle -70.12^\circ}}$$

EXERCISE 4 (Mesh analysis)



Calculate I_0

$$I_0 = -I_2$$

① $I_3 = 5\angle 0^\circ$

② KVL @ mesh 1 :

$$8I_1 + j10(I_1 - I_3) + (-j2)(I_1 - I_2) = 0$$

$$(8 + j10 - j2)I_1 + j2I_2 - j10 \times 5 = 0$$

$$(8 + j8)I_1 + j2I_2 = j50$$

③ $-(-j2)(I_1 - I_2) - (-j2)(I_3 - I_2) + 4I_2 + 20\angle 90^\circ = 0$

$$j2I_1 + (-j2 - j2 + 4)I_2 + j2 \times 5 + j20 = 0$$

$$j2I_1 + (4 - j4)I_2 = -j30$$

Solving simultaneous equations :

$$\left. \begin{aligned} (8 + j8)I_1 + j2I_2 &= j50 \quad (\times (-j2)) \\ j2I_1 + (4 - j4)I_2 &= -j30 \quad (\times (8 + j8)) \end{aligned} \right\} \begin{aligned} -j2(8 + j8)I_1 - j^2 4I_2 &= -j^2 100 \\ j2(8 + j8)I_1 + (4 - j4)(8 + j8)I_2 &= -j30(8 + j8) \end{aligned}$$

$$[4 + (4 - j4)(8 + j8)]I_2 = 100 + (-j30)(8 + j8)$$

$$(4 + 32 + j8^2 - j^2 32 + 32)I_2 = 100 - j240 + 240$$

$$68I_2 = 340 - j240$$

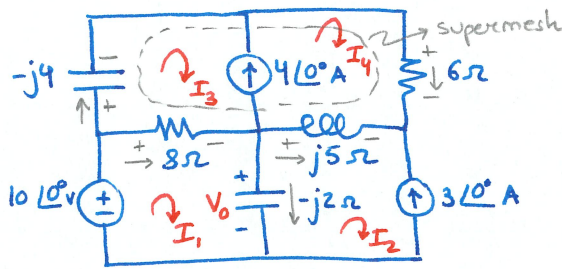
$$I_2 = \frac{340 - j240}{68} = 5 - j3.53 = 6.12 \angle -35.22^\circ$$

$$I_0 = -I_2 = -6.12 \angle -35.22^\circ = 6.12 \angle 144.78^\circ$$

$\uparrow$
 $-1 = 1 \angle 180^\circ$

EXERCISE 5 (Mesh analysis)

Find V_0



① KVL in mesh 1:

$$-10 + 8(I_1 - I_3) + (-j2)(I_1 - I_2) = 0 \Rightarrow (8 - j2)I_1 + j2I_2 - 8I_3 = 10$$

② $I_2 = -3\angle 0^\circ$

$$\Rightarrow I_2 = -3$$

③ KVL in supermesh:

$$-j4I_3 + 6I_4 - j5(I_2 - I_4) - 8(I_1 - I_3) = 0 \Rightarrow -8I_1 - j5I_2 + (8 - j4)I_3 + (6 + j5)I_4 = 0$$

④ Constraint equation

$$I_4 - I_3 = 4\angle 0^\circ \Rightarrow I_4 = 4 + I_3$$

Solve simultaneous equations:

Substitute ② in ①: $(8 - j2)I_1 - 8I_3 = 10 + j6$

Substitute ② and ④ in ③: $-8I_1 + j15 + (8 - j4)I_3 + (6 + j5)(4 + I_3) = 0$

$$-8I_1 + j15 + (8 - j4)I_3 + 24 + 6I_3 + j20 + j5I_3 = 0$$

$$-8I_1 + (8 - j4 + 6 + j5)I_3 = -j15 - 24 - j20$$

$$-8I_1 + (14 + j)I_3 = -24 - j35$$

$$\left. \begin{array}{l} (8 - j2)I_1 - 8I_3 = 10 + j6 \quad (\times (14 + j)) \\ -8I_1 + (14 + j)I_3 = -24 - j35 \quad (\times 8) \end{array} \right\} \begin{array}{l} (8 - j2)(14 + j)I_1 - 8(14 + j)I_3 = (10 + j6)(14 + j) \\ -64I_1 + 8(14 + j)I_3 = -192 - j280 \end{array}$$

$$[(8 - j2)(14 + j) - 64]I_1 = (10 + j6)(14 + j) + (-192 - j280)$$

$$(112 + 8j - j28 - \underbrace{j^2 2}_{+2} - 64)I_1 = 140 + j10 + j84 + \underbrace{j^2 6}_{-6} - 192 - j280$$

$$(50 - j20)I_1 = -58 - j186$$

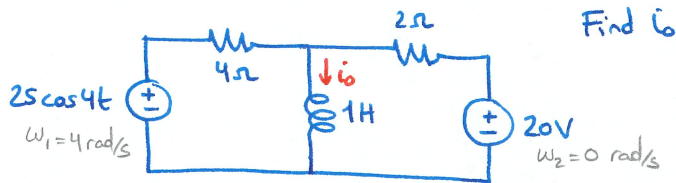
$$I_1 = \frac{-58 - j186}{50 - j20} = \frac{194.83 \angle -107.32^\circ}{53.85 \angle -21.8^\circ} = 3.618 \angle -85.52^\circ = 0.28 - j3.61$$

We can calculate V_0 once we know I_1 and I_2 :

$$V_0 = -j2(I_1 - I_2) = 2 \angle -90^\circ \times (0.28 - j3.61 + 3) = 2 \angle -90^\circ \times (3.28 - j3.61) = 2 \angle -90^\circ \times 4.88 \angle -47.74^\circ$$

$$V_0 = \underline{\underline{9.76 \angle -137.74^\circ = 9.76 \angle 222.26^\circ \text{ V}}}$$

EXERCISE 7 (superposition)



$$i_0 = i_0' + i_0''$$

① I_0' due to AC voltage source (turn off DC voltage source)



First, convert to phasor domain

$$V_t = 25 \cos 4t \rightarrow V_t = 25 \angle 0^\circ$$

$$Z_{4\Omega} = 4 \Omega$$

$$Z_{2\Omega} = 2 \Omega$$

$$Z_{1H} = j\omega L = j4 \times 1 = j4 \Omega$$



Then, calculate I_0 using voltage divider and Ohm's law Z_{eq}

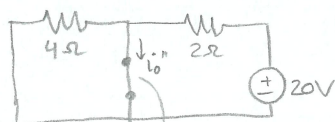
$$V_0' = 25 \angle 0^\circ \frac{Z_{eq}}{4 + Z_{eq}} \quad \text{where } Z_{eq} = 2 \parallel j4$$

$$Z_{eq} = 2 \parallel j4 = \frac{j8}{2 + j4} = 1.6 + j0.8$$

$$V_0' = 25 \angle 0^\circ \frac{1.6 + j0.8}{5.6 + j0.8} = 7.5 + j2.5 \text{ V}$$

$$I_0' = \frac{V_0'}{j4} = 1.976 \angle -71.56^\circ \text{ A} \rightarrow i_0' = 1.976 \cos(4t - 71.56^\circ)$$

② I_0'' due to DC voltage source (turn off AC voltage source)



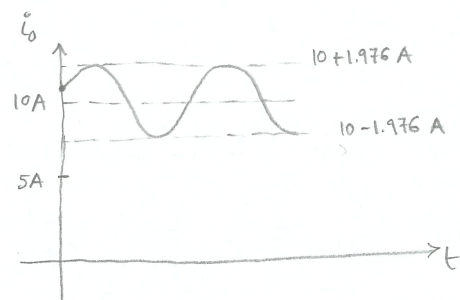
$$i_0'' = \frac{20\text{V}}{2\Omega} = 10 \text{ A}$$

inductor behaves as short circuit in DC steady state

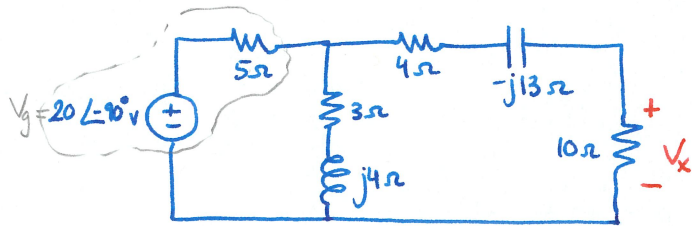
③ Final result

$$i_0 = i_0' + i_0'' = 10 + 1.976 \cos(4t - 71.56^\circ)$$

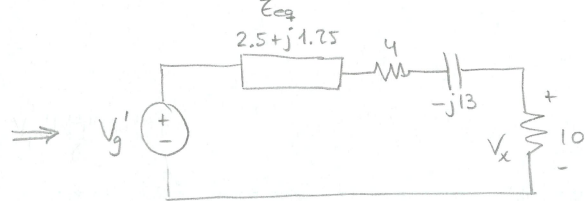
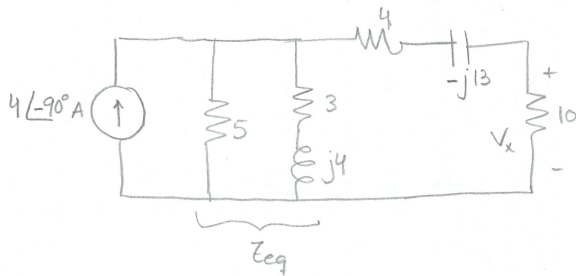
we can only add them as waveforms, not as phasors



EXERCISE 9 (Source transformation)



Calculate V_x using source transformation



$$Z_{eq} = 5 \parallel (3 + j4) = \frac{5(3 + j4)}{5 + 3 + j4} = \frac{15 + j20}{8 + j4} = \frac{25 \angle 53.13^\circ}{8.94 \angle 26.57^\circ} = 2.796 \angle 26.56^\circ = 2.5 + j1.25$$

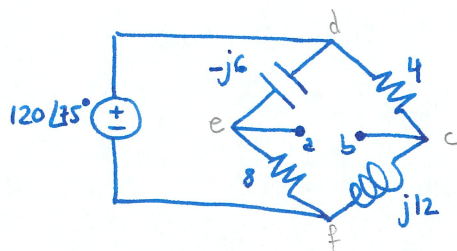
$$V_g' = 4 \angle -90^\circ \times Z_{eq} = 4 \angle -90^\circ \times 2.796 \angle 26.56^\circ = 11.184 \angle -63.44^\circ = 5 - j10$$

Use voltage division to find V_x :

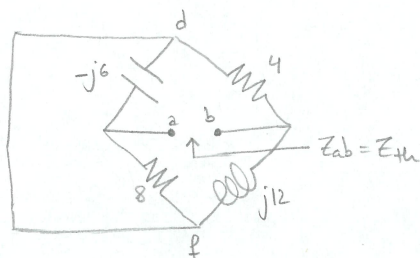
$$V_x = 11.184 \angle -63.44^\circ \frac{10}{2.5 + j1.25 + 4 - j13 + 10} = \frac{111.84 \angle -63.44^\circ}{16.5 - j11.75} = \frac{111.84 \angle -63.44^\circ}{20.26 \angle -35.46^\circ} = \underline{\underline{5.52 \angle -27.98^\circ \text{ V}}}$$

EXERCISE 11 (Thevenin equivalent)

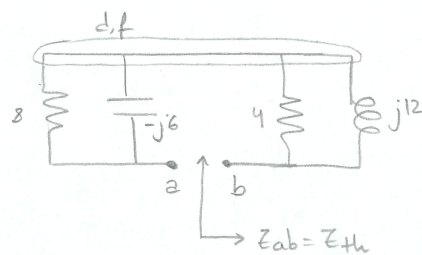
Obtain the Thevenin equivalent circuit at terminals a-b



* Find $Z_{th} = Z_{ab}$ by turning off all independent sources:



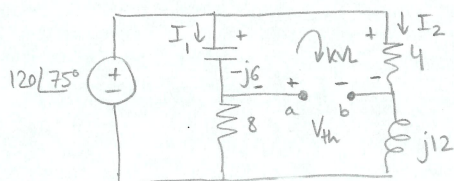
Redraw circuit



$$\begin{aligned} Z_{th} = Z_{ab} &= [8 \parallel (-j6)] + [4 \parallel (j12)] = \frac{-j48}{8-j6} + \frac{j48}{4+j12} = \\ &= 2.88 - j3.84 + 3.6 + j1.2 = \underline{\underline{6.48 - j2.64}} \end{aligned}$$

* Find $V_{th} = V_{oc}$

Multiple options. Start by finding V_{th} using KVL in upper (or lower) mesh then $\left\{ \begin{array}{l} \text{- current division} \\ \text{- voltage division} \\ \vdots \end{array} \right.$



KVL in upper mesh:

$$\textcircled{1} \quad -V_{-j6} + V_4 - V_{th} = 0 \Rightarrow V_{th} = -V_{-j6} + V_4$$

Calculate V_{-j6} and V_4 using voltage divider (or using current divider to find I_1 and I_2 and then Ohm's law)

$$V_{-j6} = 120 \angle 75^\circ \frac{-j6}{8-j6} = 66.81 + j26.82 \text{ V}$$

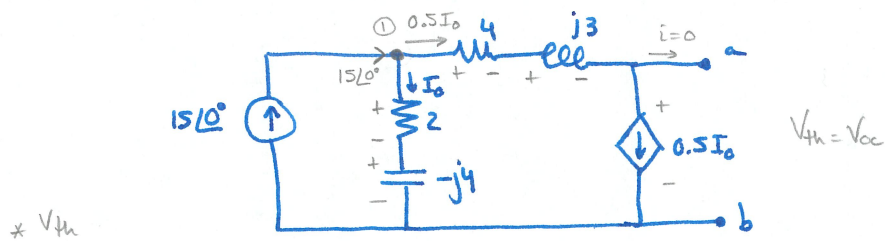
$$V_4 = 120 \angle 75^\circ \frac{4}{4+j12} = 37.88 + j2.27 \text{ V}$$

Substitute in $\textcircled{1}$:

$$V_{th} = -V_{-j6} + V_4 = -66.81 - j26.82 + 37.88 + j2.27 = -28.94 - j24.54 = \underline{\underline{37.95 \angle -139.7^\circ}}$$

EXERCISE 12 (Thevenin equivalent)

Find the Thevenin equivalent as seen from terminals a-b



KCL in node 1 to find I_0 :

$$15\angle 0^\circ = 0.5I_0 + I_0$$

$$1.5I_0 = 15\angle 0^\circ$$

$$I_0 = 10\angle 0^\circ \text{ A}$$

Use KVL in mesh 2 to find V_{th} :

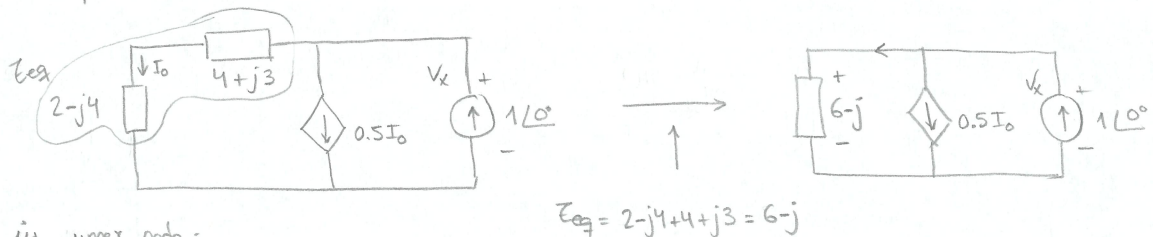
$$-(-j4)I_0 - 2I_0 + 4 \times 0.5I_0 + j3 \times 0.5I_0 + V_{th} = 0$$

$$V_{th} = -j4I_0 + 2I_0 - 2I_0 - j1.5I_0 = -j5.5I_0 = -j55 = 55\angle -90^\circ \text{ V}$$

* Z_{th}

- 2 options $\left\{ \begin{array}{l} \text{Find } I_N \text{ and calculate } Z_{th} = V_{th}/I_N, \text{ where } I_N = \text{short circuit current} \\ \text{Turn off independent sources \& connect 1A current source (or 1V voltage source) in the terminals} \end{array} \right.$

Let's use option 2 here:



KCL in upper node:

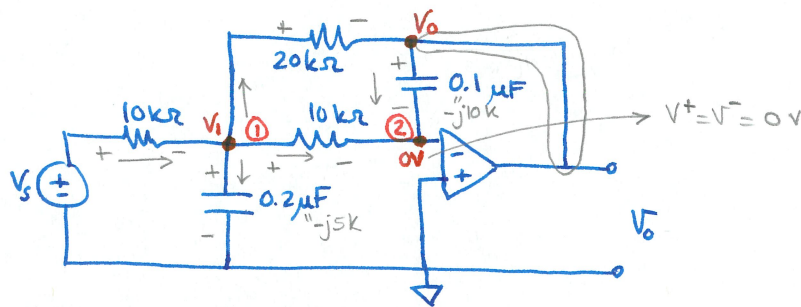
$$1\angle 0^\circ = 0.5I_0 + \frac{V_x}{6-j} \quad \text{where } I_0 = \frac{V_x}{6-j}$$

$$1\angle 0^\circ = 1.5 \frac{V_x}{6-j} \Rightarrow V_x = (6-j) \frac{1}{1.5} = 4 - j\frac{2}{3}$$

$$Z_{th} = \frac{V_x}{1\angle 0^\circ} = V_x = 4 - j\frac{2}{3} \Omega$$

EXERCISE 15 (Op Amps)

Calculate V_o if $V_s = 3 \cos(1000t)$



① Convert to phasor domain

$$V_s = 3 \cos(1000t) \rightarrow V_s = 3 \angle 0^\circ$$

$$Z_{10k} = 10k$$

$$Z_{20k} = 20k$$

$$Z_{0.2\mu F} = -j \frac{1}{\omega C_1} = -j \frac{1}{1000 \times 0.2 \times 10^{-6}} = -j5k$$

$$Z_{0.1\mu F} = -j \frac{1}{\omega C_2} = -j \frac{1}{1000 \times 0.1 \times 10^{-6}} = -j10k$$

② Use nodal analysis to find V_o as a function of V_s .

→ KCL @ node 1:

$$\frac{V_s - V_1}{10k} = \frac{V_1 - 0}{-j5k} + \frac{V_1 - 0}{10k} + \frac{V_1 - V_o}{20k} \Rightarrow 0.3 - \frac{V_1}{10} = j \frac{V_1}{5} + \frac{V_1}{10} + \frac{V_1}{20} - \frac{V_o}{20}$$

$$\left(\frac{1}{10} + j \frac{1}{5} + \frac{1}{10} + \frac{1}{20} \right) V_1 - \frac{V_o}{20} = 0.3$$

$$\left(\frac{5}{20} + j \frac{1}{5} \right) V_1 - \frac{V_o}{20} = 0.3 \quad \times 20$$

$$(5 + j4) V_1 - V_o = 6 \quad \text{①}$$

→ KCL @ node 2:

$$\frac{V_1 - 0}{10k} + \frac{V_o - 0}{-j10k} = 0 \Rightarrow \frac{V_1}{10} + j \frac{V_o}{10} = 0 \xrightarrow{\times 10} V_1 + j V_o = 0; \quad V_1 = -j V_o \quad \text{②}$$

Substitute ② in ①:

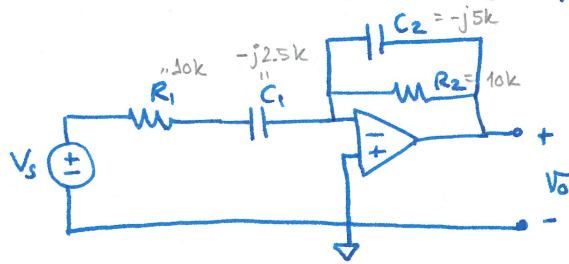
$$(5 + j4)(-j V_o) - V_o = 6; \quad -j5V_o - \overset{-1}{j^2} 4V_o - V_o = 6; \quad -j5V_o + 4V_o - V_o = 6$$

$$(3 - j5) V_o = 6$$

$$V_o = \frac{6}{3 - j5} = \frac{6 \angle 0^\circ}{5.83 \angle -59.04^\circ} = \underline{\underline{1.029 \angle 59.04^\circ}}$$

EXERCISE 17 (Op Amps)

Calculate close loop gain and phase shift if $R_1 = R_2 = 10\text{ k}$, $C_1 = 2\mu\text{F}$, $C_2 = 1\mu\text{F}$ and $\omega = 200\text{ rad/s}$



① Convert to phasor domain:

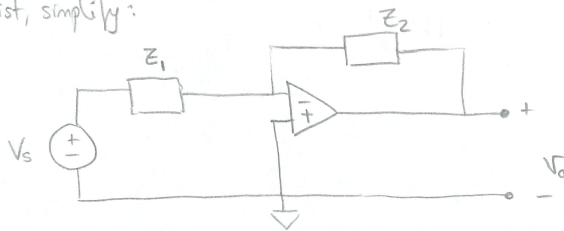
$$Z_{R_1} = Z_{R_2} = 10\text{ k}$$

$$Z_{C_1} = -j \frac{1}{\omega C_1} = -j \frac{1}{200 \times 2\mu\text{F}} = -j2.5\text{ k}$$

$$Z_{C_2} = -j \frac{1}{\omega C_2} = -j \frac{1}{200 \times 1\mu\text{F}} = -j5\text{ k}$$

② Calculate V_o as a function of V_s

First, simplify:



$$Z_1 = 10 - j2.5\text{ k} = 10.31 \angle -14.04^\circ\text{ k}$$

$$Z_2 = 10\text{ k} \parallel (-j5)\text{ k} =$$

$$= \frac{-j50}{10 - j5}\text{ k} = \frac{50 \angle -90^\circ}{11.18 \angle -26.57^\circ} = 4.47 \angle -63.43^\circ\text{ k}$$

Inverting amplifier: $V_o = -\frac{Z_2}{Z_1} V_s = -\frac{4.47 \angle -63.43^\circ}{10.31 \angle -14.04^\circ} V_s = -0.433 \angle -49.26^\circ V_s$

$$\frac{V_o}{V_s} = -0.433 \angle -49.26^\circ = \underset{\substack{\uparrow \\ -1 = 1 \angle 180^\circ}}{0.433} \angle 130.74^\circ$$